DDA5002 Optimization
Lecture 11 Convexity

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November 19, 2025

Final Exam

- **Date and Time:** December 15 (Monday), 10am–12pm
- **Topics Covered:**
  - Branch-and-Bound Algorithm
  - Optimality Conditions and KKT Conditions
  - Convexity (convex sets, convex functions, convex optimization)
  - Optimization Algorithms (Bisection, Golden Section Method, Gradient Descent, Newton's Method, Projected Gradient Method)

Outline

Outline

Karush-Kuhn-Tucker Conditions

If x is a local minimizer and if a constraint qualification (CQ) holds, then there exist λ and μ such that the following **Karush-Kuhn-Tucker (KKT) Conditions** hold:

1. Main/Stationarity Condition

$$\nabla_x L(x, \lambda, \mu) = \nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x) + \sum_{j=1}^{p} \mu_j \nabla h_j(x) = 0$$

2. Complementary Slackness

$$\lambda_i \cdot g_i(x) = 0 \quad \forall i = 1, \ldots, m$$

3. Primal Feasibility

$$g_i(x) \leq 0, \quad h_j(x) = 0 \quad \forall i = 1, \ldots, m, \ \forall j = 1, \ldots, p$$

4. Dual Feasibility

$$\lambda_i \geq 0 \quad \forall i = 1, \ldots, m$$

Constraint Qualifications

We require the collection of gradients

$$\{\nabla g_i(x) : i \in \mathcal{A}(x)\} \cup \{\nabla h_j(x) : j = 1, \ldots, p\} \tag{CQ}$$

to be **linearly independent** or to have full rank.

- This condition is a **constraint qualification (CQ)** and is called **Linear Independence Constraint Qualification (LICQ)**.
- A feasible point x satisfying the LICQ is called **regular**.
- If the local minimizer x^* does not hold a CQ, then KKT condition may not be necessary.

Motivation for Convexity

Optimality Conditions for Nonlinear (General) Optimization:

- Advantage: Conditions can be derived for very general problems.
- Disadvantage: No guarantee of global (or even local) optimality due to the generality of the conditions.

Convex Optimization:

- Convex optimization is a special but important class of optimization.
- Stationarity (FONC) is sufficient for global optimality.
- Convex optimization: local optimal is global optimal.

Outline

Collection of Vectors

weighted sum of vectors

Definition

Given a collection of vectors $x_1, \ldots x_k \in \mathbb{R}^n$:

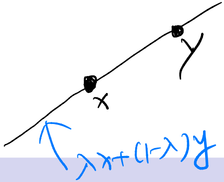
Linear combination: $\sum_{i=1}^{k} \lambda_i x_i$, $\lambda_i \in \mathbb{R}$, $\forall i$

Affine combination: $\sum_{i=1}^{k} \lambda_i x_i$, $\sum_{i=1}^{k} \lambda_i = 1$

Conic combination: $\sum_{i=1}^{k} \lambda_i x_i$, $\lambda_i \geq 0$, $\forall i$

Convex combination: $\sum_{i=1}^{k} \lambda_i x_i$, $\sum_{i=1}^{k} \lambda_i = 1, \lambda_i \geq 0, \ \forall i$

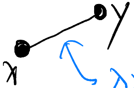
Affine Set



Definition

A set $\mathcal{X}$ is affine if $\forall x, y \in \mathcal{X}$ such that $\lambda x + (1-\lambda)y \in \mathcal{X}, \forall \lambda$

- Affine set contains the line through any two distinct points in the set.
- Example: solution set of linear equations $\{x | \underbrace{Ax = b}\}$. Conversely, every affine set can be expressed as solution set of linear equations

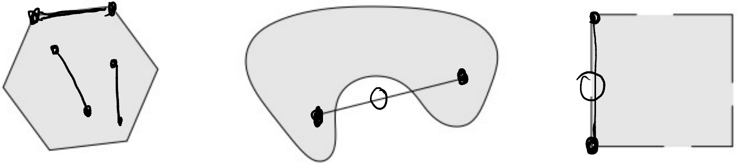


$\lambda x+(1-\lambda)y, \ \lambda \in [0,1]$

Definition

A set $\mathcal{X}$ is **convex** if $\forall x, y \in \mathcal{X}$ such that $\lambda x + (1-\lambda)y \in \mathcal{X}, \forall \lambda \in [0,1]$.

- Convex set contains line segment between any two points in the set
- Example:



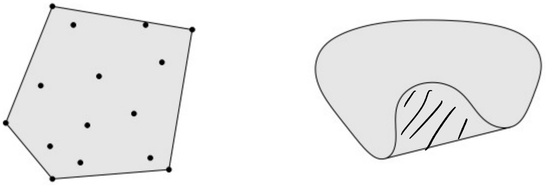
Convex | Non-Convex | Non-Convex

Convex Hull

Definition

Convex hull: set of all convex combinations of elements in the set

$$conv(\mathcal{X}) = \{\sum_i \lambda_i x_i, x_i \in \mathcal{X}, \lambda_i \in [0,1], \sum_i \lambda_i = 1\}$$

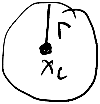


The convex hull of the set X is the smallest convex set containing all elements in X.

Convex Set Examples

1. - **Hyperplane:** set of the form $\{x \mid a^T x = b\}$ where $a \neq 0$
2. - **Halfspace:** set of the form $\{x \mid a^T x \leq b\}$ where $a \neq 0$
3. - **(Euclidean) ball** with center x_c and radius r:

$$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\}$$



  where $\|\cdot\|_2$ denotes the Euclidean norm
4 - **Polyhedron**: set of the form $\{x \mid a^T x \leq b\}$ where $a \neq 0$

1. Hyperplane : $X = \{x : a^T x = b\}$

Proof: $\forall x, y \in X$, $\forall \lambda \cancel{\in [0,1]}$

$$a^T(\lambda x + (1-\lambda) y) = \lambda a^T x + (1-\lambda) a^T y$$

$$= \lambda b + (1-\lambda) b$$

$$= b$$

So $\lambda x + (1-\lambda) y \in X$, $\forall \lambda \cancel{\in [0,1]}$

Hyperplane is an affine set and convex set.

2. Halfspace: $X = \{x : a^T x \le b\}$

Proof: $\forall x, y \in X$, $\forall \lambda \in [0,1]$

$$a^T(\lambda x + (1-\lambda) y) = \lambda a^T x + (1-\lambda) a^T y$$

$$\le \lambda b + (1-\lambda) b$$

$$= b$$

So $\lambda x + (1-\lambda) y \in X$ for $\lambda \in [0,1]$

Halfspace / Polyhedron is a convex set.

center ↓ ↙ radius

$B(x_c, r)$

3. Ball : $B = \{x : \|x - x_c\|_2 \le r\}$

Proof: $\forall x, y \in B$, $\forall \lambda \in [0,1]$

Triangle Inequality
$\|a+b\|_p \le \|a\|_p + \|b\|_p$ for $p \ge 1$

$$\| \lambda x + (1-\lambda) y - x_c \|_2$$

$$= \| \lambda (x - x_c) + (1-\lambda)(y - x_c) \|_2$$

triangle inequality →

$$\le \| \lambda (x - x_c) \|_2 + \| (1-\lambda)(y - x_c) \|_2$$

$$= \lambda \underbrace{\|x - x_c\|_2}_{\le r} + (1-\lambda) \underbrace{\|y - x_c\|_2}_{\le r}$$

$$\le \lambda r + (1-\lambda) r$$

$$= r$$

So $\lambda x + (1-\lambda) y \in B$ for $\lambda \in [0,1]$

Ball is a convex set.

Operations that Preserve Convexity I

- **Intersection:** Given a family of convex sets $\{X_a\}_{a \in A}$, the set

$$X = \bigcap_{a \in A} X_a$$

is convex.

- **Sum:** Given a family of convex sets $\{X_a\}_{a \in A}$, the set

$$X = \left\{ \sum_{a \in A} x_a : x_a \in X_a \ \forall a \in A \right\}$$

is convex.

Operations that Preserve Convexity II

- **Affine image:** If $X \subseteq \mathbb{R}^n$ is a convex set, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, then
$$AX + b = \{Ax + b : x \in X\}$$
is a convex set in $\mathbb{R}^m$.
- **Inverse affine image:** If $Y \subseteq \mathbb{R}^m$ is a convex set, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, then
$$S = \{x \in \mathbb{R}^n : Ax + b \in Y\}$$
is a convex set.

Outline

Convex Function

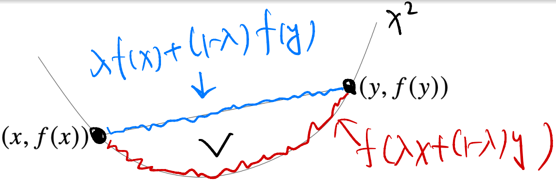
Definition

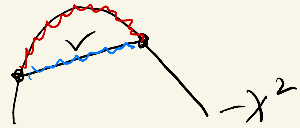
A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **convex** function iff the dom(f) is a convex set and

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for all $x, y \in \text{dom}(f)$ and $\lambda \in [0,1]$.
f is **concave** if $-f$ is convex.
f is **strictly** convex iff the dom(f) is a convex set and

$$f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda)f(y)$$





Concave:

$$f(\lambda x + (1-\lambda) y) \geq \lambda f(x) + (1-\lambda) f(y)$$

Thm: $f(\lambda x + (1-\lambda) y) = \lambda f(x) + (1-\lambda) f(y)$

$$\Updownarrow$$

$f(x)$ is a linear function.

- A Function is both convex and concave.

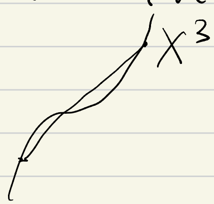
$$\Updownarrow$$

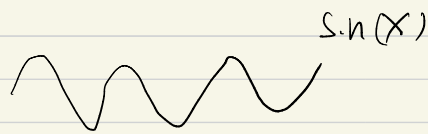
This function is a linear function.

Remark:

A function is either convex or concave. ✗ WRONG!

Non-convex functions: not convex





A set is either convex or non-convex.

No concave set.

Convex Function Examples

Convex

- affine: $ax + b$ on $\mathbb{R}$, for any $a, b \in \mathbb{R}$
- exponential: e^{ax}, for any $a \in \mathbb{R}$
- powers: x^α on $x > 0$, for $\alpha \geq 1$ or $\alpha \leq 0$
- powers of absolute value: $|x|^p$ on $\mathbb{R}$, for $p \geq 1$
- negative entropy: $x \log x$ on $x > 0$
- norms: $\|x\|_p = \underline{\left(\sum_{i=1}^n |x_i|^p\right)^{1/p}}$ for $\underline{p \geq 1}$; $\|x\|_\infty = \max_k \{x_k\}$

Concave

- affine: $ax + b$ on $\mathbb{R}$, for any $a, b \in \mathbb{R}$
- powers: x^α on $x > 0$, for $0 \leq \alpha \leq 1$ $\sqrt{x}$
- logarithm: $\log x$ on $x > 0$

Statement: Norm function is convex. X

$\|X\|_p$ is convex function if $P \geq 1$.

Proof: $1 \leq P < \infty$, $f(x) = \|X\|_p$

$\forall x, y, \quad \forall \lambda \in [0, 1]$

$$f(\lambda x + (1-\lambda) y) = \|\lambda x + (1-\lambda) y\|_p$$

triangle inequality

$$\leq \|\lambda x\|_p + \|(1-\lambda) y\|_p$$

$$= \lambda \|X\|_p + (1-\lambda) \|y\|_p$$

$$= \lambda f(x) + (1-\lambda) f(y)$$

So $\|x\|_p$ is convex for $1 \leq P < \infty$

$X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$ $\quad \|X\|_\infty = \max_i \{X_i\}$

Prove $f(x) = \max_i \{X_i\}$ is a convex function

$\forall x, y, \quad \forall \lambda \in [0, 1]$

$$f(\lambda x + (1-\lambda) y) = \max_i \{\lambda X_i + (1-\lambda) y_i\}$$

$$\leq \lambda \max_i \{X_i\} + (1-\lambda) \max_i y_i$$

$$= \lambda f(x) + (1-\lambda) f(y)$$

$\lambda \max_i \{X_i\} + (1-\lambda) \max_i \{y_i\}$

$\geq \lambda X_i + (1-\lambda) y_i \quad \forall i$

select i s.t. $\lambda X_i + (1-\lambda) y_i$ is max

$f(x) = \|x\|_0 = \#$ of nonzero elements in x.

is not convex.

Quiz: $f(x) = [x]_2$

$= 2^{nd}$ largest element of x

$f(x)$ is nonconvex.

$$\boxed{\begin{array}{l}\text{Hwo to Prove } f(x) \text{ is not convex?} \\ \exists x, y,\ \exists \lambda \in [0,1] \\ f(\lambda x + (1-\lambda) y) > \lambda f(x) + (1-\lambda) f(y)\end{array}}$$

$x = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$ $\quad y = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$ $\quad \lambda = \frac{1}{2}$

$f(x) = 2$ $\quad f(y) = 2$

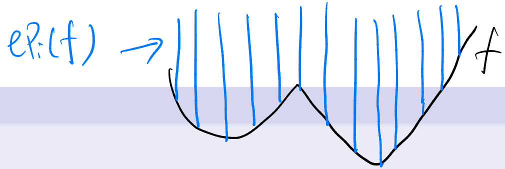
$f(\lambda x + (1-\lambda) y) = f\left(\frac{1}{2}\begin{bmatrix} 10 \\ 2 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 2 \\ 10 \end{bmatrix}\right) = f\left(\begin{bmatrix} 6 \\ 6 \end{bmatrix}\right) = 6$

$\lambda f(x) + (1-\lambda) f(y) = \frac{1}{2} \times 2 + \frac{1}{2} \times 2 = 2$

So $f(x)$ is not convex.

Remark: $f(x)$ is not concave function.

Examples: Epigraph and Sublevel Set



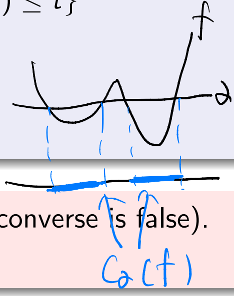
Definition

The **epigraph** of $f : \mathbb{R}^n \to \mathbb{R}$ is

$$epi(f) = \{(x, t) \in \mathbb{R}^{n+1} \mid x \in \text{dom}(f), f(x) \le t\}$$

The α-**sublevel set** of $f : \mathbb{R}^n \to \mathbb{R}$ is

$$C_\alpha(f) = \{x \in \text{dom}(f) \mid f(x) \le \alpha\}$$



- The sublevel sets of convex functions are convex (converse is false).
- f is convex iff $epi(f)$ is a convex set.

If f is convex (function), then $C_a(f)$ is convex (set).

Proof: $\forall x, y \in C_a(f)$, $\forall \lambda \in [0,1]$

$\hookrightarrow f(x) \leq a$, $f(y) \leq a$

Since f is convex,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

$$\leq \lambda a + (1-\lambda) a$$

$$= a$$

$\Rightarrow \lambda x + (1-\lambda) y \in C_a(f)$ for $\lambda \in [0,1]$

$\Rightarrow C_a(f)$ is convex.

Remark: If $C_a(f)$ is convex, then f may not be convex

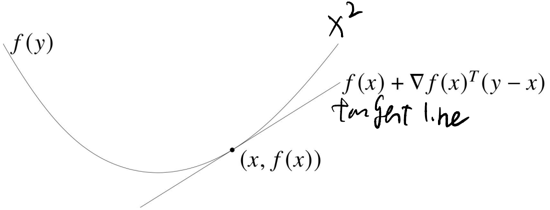
First-order Condition of Convex Functions

First-order condition of convex functions

A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ with convex domain is convex iff

$$f(y) \geq \underbrace{f(x) + \nabla f(x)^\mathsf{T}(y - x)}_{\text{tangent line}}$$

for all $x, y \in \mathrm{dom}(f)$.



Second-order Condition of Convex Functions

Second-order condition of convex functions

A twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ with convex domain is convex iff

$f''(x) \geq 0$

$$\nabla^2 f(x) \succcurlyeq 0 \quad \text{(the Hessian is positive semi-definite)}$$

$[\nabla^2 f(x)]_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$

for all $x \in \mathrm{dom}(f)$.

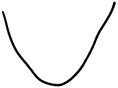
Positive Semi-definite

A $n \times n$ matrix A is positive semi-definite if $x^T A x \geq 0$ for all $x \in \mathbb{R}^n$. Equivalently, A is positive semi-definite if all its eigenvalues are nonnegative.

Remark: Concave function has a negative semi-definite Hessian.

Examples of Using First and Second-order Conditions

$P \succeq 0$



$\nabla f(x) = Px + q$

$\nabla^2 f(x) = P$

$f(x)$ is convex if $P \succeq 0$

- Quadratic function: $f(x) = (1/2)x^\intercal P x + q^\intercal x + r = \frac{1}{2}Px^2 + qx + r$
- Least squares objective: $f(x) = ||Ax - b||_2^2 = (Ax-b)^2$

$\nabla f(x) = 2A^T(Ax-b) = 2A^2x - 2Ab$

$\nabla^2 f(x) = 2A^TA \succeq 0$

$f(x)$ is convex for any A.

Convex Function Calculus I

non-negative weighted sum

Lemma: Sum Rule

If $a_1, \ldots, a_m \geq 0$, and $f_1, \ldots, f_m$ are convex (concave) functions, then $f = a_1 f_1 + \cdots + a_m f_m$ is a convex (concave) function.

- Nonnegative linear combination preserves convexity (concavity).
- Examples: $x_1^2 + x_2^2$, $e^x + |x|$.

2. **Lemma: Composition with Linear Functions**

If $g : \mathbb{R}^m \to \mathbb{R}$ is convex (concave) and $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ are given, then $f(x) := g(Ax + b)$ is convex (concave).

- Examples: e^{2x+3} (convex), $(x_1 - x_2)^2 + (x_2 + x_3)^2$ (convex), $\|Ax - b\|$ (convex), $\log(-2x_1 + 3x_2 + 5)$ (concave).

2. Proof: $\forall x, y, \ \forall \lambda \in [0,1]$

$$f(\lambda x+(1-\lambda)y) \underset{\text{def of } f}{=} g(A(\lambda x+(1-\lambda)y)+b)$$

$$= g(\lambda Ax+(1-\lambda)Ay+b)$$

$$\overset{b=\lambda b+(1-\lambda)b}{=} g(\lambda Ax+\lambda b+(1-\lambda)Ay+(1-\lambda)b)$$

$$= g(\lambda(Ax+b)+(1-\lambda)(Ay+b))$$

$$\overset{g \text{ is convex}}{\leq} \lambda g(Ax+b)+(1-\lambda)g(Ay+b)$$

$$= \lambda f(x)+(1-\lambda)f(y)$$

Convex Function Calculus II

Lemma: Pointwise Maximum
If $f_1, \ldots, f_m$ are convex functions, then

$$f(x) = \max\{f_1(x), \ldots, f_m(x)\}$$

is a convex function (this can be extended to uncountably many).

- Examples: $|x| = \max\{-x, x\}$, $\max_i\{a_i^\top \mathbf{x} + b_i\}$, $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$.

Lemma: Pointwise Minimum
If $f_1, \ldots, f_m$ are concave functions, then

$$f(x) = \min\{f_1(x), \ldots, f_m(x)\}$$

is a concave function (this can be extended to uncountably many).

- Examples: $-|x| = \min\{-x, x\}$, $\min_i\{a_i^\top \mathbf{x} + b_i\}$.

Convex Function Calculus III

$$g(x_1, \cdots, x_m)$$

$x_1 \uparrow \rightarrow g \uparrow$

$\vdots$

$x_m \uparrow \rightarrow g \uparrow$

Lemma: Composition with Convex and Nondecreasing Function

If $g : \mathbb{R}^m \to \mathbb{R}$ is convex and componentwise nondecreasing, and each $f_i : \mathbb{R}^n \to \mathbb{R}$ is convex, then

$$f(x) = g(f_1(x), \ldots, f_m(x))$$

is a convex function.

- Examples: $f(x) = e^{\sum_{i=1}^{m} |a_i x - b_i|}$.

$$f(x) = e^{\sum_i |a_i x - b_i|}$$

$l_i(x) = a_i x - b_i \rightarrow$ linear $\rightarrow$ convex

$h_i(x) = |l_i| = |a_i x - b_i|$ convex by Max Rule

$= \max\{a_i x - b_i, -a_i x + b_i\}$

$g(x) = \sum_i h_i(x) = \sum_i |a_i x - b_i| \rightarrow$ convex by Sum Rule

$f(x) = e^{g(x)} = m(g(x)) \rightarrow$ convex by composition Rule

$m(x) = e^x$ convex & non-decreasing

Convex Function in ML Examples

- Linear Regression

$$f(\beta)$$

$$\min_{\beta \in \mathbb{R}^n} \sum_{i=1}^{m} \left(x_i^\top \beta - y_i \right)^2 = \|X\beta - \mathbf{y}\|_2^2$$

- Robust Regression

$$\min_{\beta \in \mathbb{R}^n} \sum_{i=1}^{m} \left| x_i^\top \beta - y_i \right| = \|X\beta - \mathbf{y}\|_1$$

- Logistic Regression

$$f(w) = \frac{1}{m} \sum_i g_i(w) \rightarrow \text{Sum Rule}$$

Convex

$$\min_{w \in \mathbb{R}^n} \frac{1}{m} \sum_{i=1}^{m} \underbrace{\log \left(1 + \exp(-y_i \cdot \mathbf{w}^\top x_i) \right)}_{g_i(w)}$$

1. $f(\beta) = \|X\beta - y\|_2^2$ Convex

$\nabla f(\beta) = 2X^T(X\beta - y)$

$\nabla^2 f(\beta) = 2X^T X \succeq 0$

2. $f(\beta) = \|X\beta - y\|_1$

$$= \underbrace{\sum_i \underbrace{| \underbrace{x_i^T\beta - y_i}_{\text{linear}} |}_{\text{convex by max rule}}}_{\text{convex by sum rule}}$$

3. $g(z) = \ln(1 + \exp(z))$ $\quad z = -y_i w^T x_i$ linear

$$g'(z) = \frac{e^z}{1+e^z}$$

$$g''(z) = \frac{e^z(1+e^z) - e^z \cdot e^z}{(1+e^z)^2} = \frac{\overset{\geq 0}{e^z}}{\underset{>0}{(1+e^z)^2}} \geq 0$$

$g(z)$ is convex in z.

Max likelihood $\Longleftrightarrow$ Max log-likelihood

max. $L(\theta) = \prod_i f_i$

$\Updownarrow$

max $\log L(\theta) = \log \prod_i f_i = \sum_i \log f_i$

concave

Outline

Convex optimization

max. $g(x)$ → concave function
s.t. $x \in X$ → convex set

$$\begin{aligned} \min. \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

→ convex function
→ convex set

If f is a convex function and X is a convex set, then the problem is a convex optimization problem. (Minimize a convex function is equivalent to maximize a concave function.) Otherwise, we call it non-convex optimization problem

Examples:

- o $\min\{x^2 : -1 \le x \le 1\}$ is a convex opt. problem.
- o $\min\{-x^2 : -1 \le x \le 1\}$ is not a convex opt. problem. concave
- o $\min\{x^2 : \underbrace{-5 \le x \le 5, x \in \mathbb{Z}}_{\text{not convex set}}\}$ is not a convex opt. problem.

Constraint Types

What constraints would make the feasible region convex?

Convex Constraint

Let f be a convex (concave) function. Then for any number c, the set $\Gamma_c = \{x : f(x) \leq (\geq) c\}$ is a convex set. $\hookrightarrow$ α-sub level set

Therefore,

- If we have constraint $g(x) \leq 0$, and $g(x)$ is convex, then this is a convex constraint.
- If we have constraint $g(x) \geq 0$, and $g(x)$ is concave, then this is a convex constraint.
- Linear constraints are always convex constraints.
- Sometimes, even if a constraint doesn't appear to be in the above form, it still could be a convex constraint.

Being able to identify convex problem is an important skill.

Recognizing Convex Optimization Problem

$$\begin{aligned} \min. \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

$\begin{cases} h_j(x) \le d_j & \text{Convex} \\ h_j(x) \ge d_j \Leftrightarrow -h_j(x) \le -d_j & \text{Concave, Convex} \end{cases}$

Suppose X is in the following format

$$X = \{\underline{g_i(x) \leq b_i} \ \ \forall i \in I, \ \ \underline{h_j(x) = d_j} \ \ \forall j \in J\},$$

if f is convex, g_i is convex for all $i \in I$ and h_j is ~~affine~~ for all $j \in J$, then the problem is a convex optimization problem.

linear

Sufficient not necessary

Checking Convexity

- Check that all variables are continuous ✓
- Check that the objective function is convex ✓
- Check each equality constraint to see if it is linear $g(x) \le b$
- Write each constraint as an inequality constraint in $\leq$ form with a constant on the right-hand-side, and check the convexity of the function on the left-hand-side

If it passes all the checks then you have a convex optimization problem. Otherwise, it may or may not be convex (the conditions are sufficient, not necessary).

Convex Optimization Example

$$\begin{aligned} \min. \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & \frac{x_1}{1+x_2^2} \le 0 \\ & (x_1 + x_2)^2 = 0 \end{aligned}$$

Convex ← Sum Rule

not convex function

not linear

the problem is equivalent to

$$\begin{aligned} \min. \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & x_1 \le 0 \\ & x_1 + x_2 = 0 \end{aligned}$$

Convex OPT

Polyhedron

Example

Commen trick in ML

max. $\log xyz$

$\Updownarrow$ max $\underbrace{\log x + \log y + \log z}_{\text{concave}}$

$$
\begin{aligned}
\text{max.} \quad & xyz \\
\text{s.t.} \quad & z^2 - xy \le 0 \\
& x, y, z \ge 0
\end{aligned}
$$

$$g(x,y,z) \preceq 0$$

$$X = \{ (x,y,z) : z^2 - xy \leq 0, \; x, y, z \geq 0 \}$$

$g(x,y,z) = z^2 - xy$ Convex?

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial x} \\ \frac{\partial g}{\partial y} \\ \frac{\partial g}{\partial z} \end{bmatrix} = \begin{bmatrix} -y \\ -x \\ 2z \end{bmatrix}$$

Hessian

$$\nabla^2 g = \begin{array}{c|ccc} & x & y & z \\ \hline x & 0 & -1 & 0 \\ y & -1 & 0 & 0 \\ z & 0 & 0 & 2 \end{array}$$

Eigenvalues of $\nabla^2 g$ are $-1, 1, 2$

$\nabla^2 g$ is not positive semi-definite

$g(x,y,z)$ is not convex.

not convex

$$\mathcal{X} = \{(x,y,z) : z^2 - xy \le 0,\ x, y, z \ge 0\}$$

Prove $\mathcal{X}$ is a convex set.

$\forall (x_1, y_1, z_1)$ and $(x_2, y_2, z_2) \in \mathcal{X} \Rightarrow z_1^2 \le x_1 y_1$, $z_2^2 \le x_2 y_2$

$\forall \lambda \in [0,1]$

$$\left(\underset{\substack{\vee\!\mid \\ 0}}{\lambda x_1 + (1-\lambda)x_2},\ \underset{\substack{\vee\!\mid \\ 0}}{\lambda y_1 + (1-\lambda) y_2},\ \underset{\substack{\vee\!\mid \\ 0}}{\lambda z_1 + (1-\lambda) z_2}\right)$$

$$\begin{aligned}
&(\lambda z_1 + (1-\lambda) z_2)^2 \\
&= \lambda^2 z_1^2 + (1-\lambda)^2 z_2^2 + 2\lambda(1-\lambda) z_1 z_2 \\
&\le \lambda^2 x_1 y_1 + (1-\lambda)^2 x_2 y_2 + \lambda(1-\lambda)(x_1 y_2 + x_2 y_1) \\
&= \lambda^2 x_1 y_1 + (1-\lambda)^2 x_2 y_2 + \lambda(1-\lambda) x_1 y_2 + \lambda(1-\lambda) x_2 y_1 \\
&= (\lambda x_1 + (1-\lambda) x_2) \cdot (\lambda y_1 + (1-\lambda) y_2)
\end{aligned}$$

$$\boxed{\begin{aligned}
&a + b \ge 2\sqrt{ab} \\
&z_1^2 \le x_1 y_1 \\
&z_2^2 \le x_2 y_2 \\
&2\sqrt{z_1^2 z_2^2} = 2 z_1 z_2 \\
&\le 2\sqrt{x_1 y_1}\sqrt{x_2 y_2} \\
&= 2\sqrt{(x_1 y_2)(x_2 y_1)} \\
&\le x_1 y_2 + x_2 y_1
\end{aligned}}$$

So $\mathcal{X}$ is a convex set

Remark: If $g(x)$ is not convex,

$\{x : g(x) \le b\}$ may still a convex set.

Convex Optimization: Local is Global

$$\text{(P)}: \quad \min . \{f(x): \; x \in X\}$$

Theorem

If (P) is a convex optimization problem then a solution $x^* \in X$ is a local optimal solution of (P) iff it is a global optimal solution.

Proof:

Suppose x^* is a local optimal not global

$$\exists\, x' \in \mathcal{X} \quad \text{s.t.} \quad f(x') < f(x^*)$$

Since $f(x)$ is convex, $\forall \lambda \in [0,1]$

$x = x' \in \mathcal{X}$
$y = x^* \in \mathcal{X}$

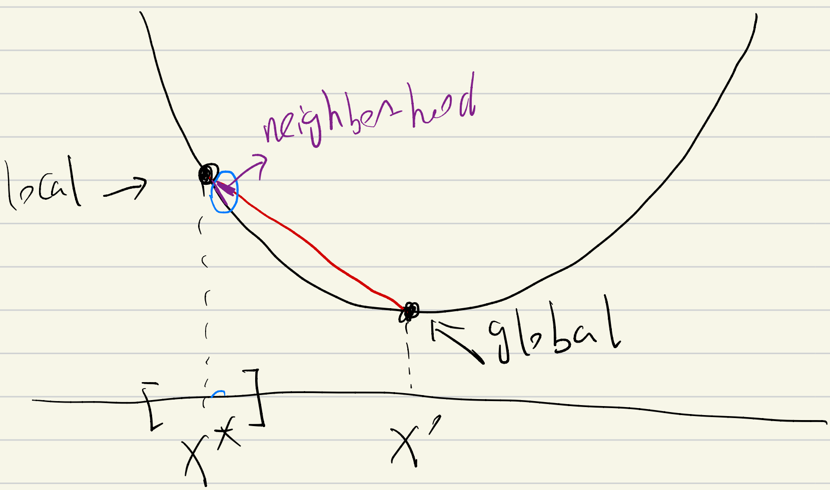
$$f(\lambda x' + (1-\lambda)x^*) \le \lambda f(x') + (1-\lambda) f(x^*)$$

λ small, $\lambda x' + (1-\lambda)x^*$ is in the neighbor-hood of x^*

$$< \lambda f(x^*) + (1-\lambda) f(x^*)$$

$$= f(x^*)$$

This is a contradiction of x^* is local optimal



Stationarity and Global Optimality: Unconstrained Case

Theorem: Stationarity & Global Optimality unconstrained

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at x^*. Then, x^* is a global minimum of $\min_{x \in \mathbb{R}^n} f(x)$ if and only if

$$\nabla f(x^*) = 0. \Rightarrow x^* \text{ is global min}$$

Proof by first-order condition

Remarks:

- In unconstrained convex optimization, stationarity (FONC) is both sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problem provides global optimal solutions.
- For the least squares problem $\min_\beta \|X\beta - y\|^2$, the FONC is convex

$$X^\top X\beta = X^\top y. \qquad \hat{\beta} = (X^T X)^{-1} X^T y$$

All such β are global minimizers.

The ‘Easy’ and ‘Difficult’ Optimization Problems

- Linear v.s. nonlinear?
- Differentiable v.s. nondifferentiable?

Classify whether a problem is hard or easy: **Convex (easy)** v.s. **nonconvex (hard)**.

- **Convex optimization:** Benign global geometry and hence reasonable algorithms can almost always find the global minimum.
- **Nonconvex optimization:** Can be anything (e.g., the *saddle point*). In general, finding a stationary point is NP-hard.

Outline

Lagrangian Dual Problem

Given original problem (P):

$$v_P = \min_x \{f(x) : g_i(x) \le b_i, \ \forall i \in I, \ h_j(x) = d_j, \ \forall j \in J\}$$

The dual function is defined as: → Lagrangian Relaxation

$$\begin{aligned}
\mathcal{L}(\lambda, \mu) &= \min_x L(x, \lambda, \mu) \\
&= \min_x \left\{ \overbrace{f(x) + \sum_{i \in I} \lambda_i [g_i(x) - b_i] + \sum_{j \in J} \mu_j [h_j(x) - d_j]}^{L(x, \lambda, \mu) \, =} \right\}.
\end{aligned}$$

Dual function provides a lower bound: $v_P \ge \mathcal{L}(\lambda, \mu)$ for all $\lambda \ge 0$ and μ.
The best lower bound is given by solving the dual problem:

$$(D): \quad v_D = \max_{\lambda, \mu} \{\mathcal{L}(\lambda, \mu) : \lambda \ge 0\}$$

Duality for General Optimization Problems

Now we have the dual problem

$$\max_{\lambda \geq 0, \mu} \quad \{\min_x L(x, \lambda, \mu)\}$$

- λ and μ are the dual variables.
- If the inner problem has an explicit form, then the dual problem is an explicit maximization problem in λ and μ (this is the case for linear optimization problems).
- However, in general, the inner problem may not have an explicit solution. In those cases, the dual problem doesn't have an explicit form (but it is still an optimization problem for λ and μ).

Dual is Convex Optimization

Remember the dual problem is:

Concave

$L(x,\lambda,\mu)$

$$\max_{\lambda \geq 0, \mu} \quad \{\min_x \underbrace{L(x, \lambda, \mu)}_{\text{linear}}\}$$

where L is a linear function of λ and μ.

- Therefore, $\min_x L(x, \lambda, \mu)$ is always concave in λ and μ.
- Thus the dual problem is always a convex optimization problem — regardless of whether the primal problem is a convex optimization problem or not.
- The dual of dual problem does not necessarily equal to the primal problem.

Example

Consider the following problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{x}^T\boldsymbol{x} \\ \text{subject to} \quad & A\boldsymbol{x} \le \boldsymbol{b} \end{aligned}$$

It is a nonlinear optimization problem. Now let's construct its dual problem.

Weak Duality

Theorem (Weak Duality)

If the primal and dual optimal value is v_P and v_D, then

$$v_D \leq v_P$$

Note that this always holds (even if the primal problem is nonconvex).

Duality Gap $= V_P - V_D \geq 0$

Strong Duality

$C^T x = b^T y$

For LP, we have the strong duality theorem, saying that the optimal values of primal and dual problems are always the same.

Unfortunately, this is not necessarily true for nonlinear optimization problems. We call the difference between the primal optimal value and dual optimal value the duality gap.

- For LP, the duality gap is always 0.
- For general optimization problems, there could be a positive duality gap. $V_P - V_D > 0$
- Again, by the weak duality theorem, it must be that the optimal value of the minimization problem is larger than that of the maximization problem.

Strong duality : duality gap $= 0$

$V_P = V_D$

Strong Duality Theorem

Convex OPT + Slater's $\Longrightarrow$ $V_P = V_D$

Strong duality

Theorem (Strong Duality Theorem)

If the primal problem is a convex optimization problem and also Slater's condition holds, then the duality gap is 0, i.e., $v_D = v_P$.

Definition (Slater's Condition)

$g_i(x) \le b_i$

If the original problem (P) is convex (i.e., f and g_i are convex for all $i \in I$, and h_j are affine for all $j \in J$), and there exists a strictly feasible $x \in \mathbb{R}^n$ such that:

$$g_i(x) < b_i \quad \forall i \in I, \qquad h_j(x) = d_j \quad \forall j \in J,$$

then Slater's Condition holds and strong duality holds.

- In other words, Slater's condition requires that there is a strict interior solution for nonlinear constraints.

Outline

General Optimality Conditions

Now, We consider the following nonlinear program:

$$\begin{aligned}
\min_{x \in \mathbb{R}^n} \quad & f(x) \\
\text{subject to} \quad & g_i(x) \leq 0, \quad \forall i = 1, \ldots, m, \\
& h_j(x) = 0, \quad \forall j = 1, \ldots, p.
\end{aligned}$$

Karush-Kuhn-Tucker Conditions

Karush-Kuhn-Tucker (KKT) Conditions:

- Main/Stationarity Condition

$$\nabla_x L(x, \lambda, \mu) = \nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x) + \sum_{j=1}^{p} \mu_j \nabla h_j(x) = 0$$

- Complementary Slackness

$$\lambda_i \cdot g_i(x) = 0 \quad \forall i = 1, \ldots, m$$

- Primal Feasibility

$$g_i(x) \le 0, \quad h_j(x) = 0 \quad \forall i = 1, \ldots, m, \ \forall j = 1, \ldots, p$$

- Dual Feasibility

$$\lambda_i \ge 0 \quad \forall i = 1, \ldots, m$$

KKT Conditions Theory for Convex Porblems

Theorem

1. *For convex optimization problem with strong duality holds (e.g., Slater's condition holds), if* $x^\star$ *and* $\lambda^\star, \mu^\star$ *are primal and dual optimal solutions, then* $x^\star$ *and* $\lambda^\star, \mu^\star$ *satisfy the KKT conditions.*
2. *For convex optimization problem, if* $x^\star$ *and* $\lambda^\star, \mu^\star$ *satisfy the KKT conditions, then* $x^\star$ *and* $\lambda^\star, \mu^\star$ *are primal and dual optimal solutions and strong duality holds.*

- Convex OPT + Strong duality (Slater's)

  KKT $\Leftrightarrow$ optimal sols

- Convex OPT

  KKT $\Rightarrow$ strong duality + optimal soln

- General OPT + CQ

  local optimal $\Rightarrow$ KKT